

# ■ ProspectIQ

## Tools & Technologies Reference

XLVentures.AI Hackathon 2025

### 1. Platform Overview

ProspectIQ is a multi-agent AI platform for automated B2B customer discovery. It is built on a modern Python stack combining a large language model (LLM) inference API, a real-time web search API, a multi-agent orchestration framework, a REST API backend, and an interactive frontend dashboard. All components are open-source or free-tier SaaS.

### 2. LLM & Inference

#### Groq API

|            |                                                                                       |
|------------|---------------------------------------------------------------------------------------|
| Role       | LLM inference provider — hosts and serves the language model                          |
| Free Tier  | 500,000 tokens/day (llama-3.1-8b-instant)                                             |
| Why chosen | Fastest open-source inference available; free tier sufficient for hackathon workloads |
| Website    | <a href="https://console.groq.com">https://console.groq.com</a>                       |

#### LLaMA 3.1 8B Instant

|            |                                                                                             |
|------------|---------------------------------------------------------------------------------------------|
| Role       | Core language model powering all 7 AI agents                                                |
| Parameters | 8 billion                                                                                   |
| Context    | 128k tokens                                                                                 |
| Provider   | Meta (served via Groq)                                                                      |
| Why chosen | Fast, capable, and free; ideal for structured output generation across multiple agent calls |

### 3. Agent Orchestration

#### LangGraph

|            |                                                                                        |
|------------|----------------------------------------------------------------------------------------|
| Version    | 0.4.8                                                                                  |
| Role       | Multi-agent pipeline orchestration using a directed StateGraph                         |
| Pattern    | Sequential 7-node graph; each node is an agent function that reads/writes shared state |
| State type | ProcureAIState (TypedDict) — typed shared memory across all agents                     |
| Why chosen | Purpose-built for stateful multi-agent workflows; clean separation of agent logic      |

|                |                                                                                                   |
|----------------|---------------------------------------------------------------------------------------------------|
| <b>Website</b> | <a href="https://langchain-ai.github.io/langgraph/">https://langchain-ai.github.io/langgraph/</a> |
|----------------|---------------------------------------------------------------------------------------------------|

## LangChain

|                |                                                                           |
|----------------|---------------------------------------------------------------------------|
| <b>Version</b> | 0.3.25                                                                    |
| <b>Role</b>    | LLM integration layer; provides ChatGroq wrapper                          |
| <b>Usage</b>   | <code>llm.invoke(prompt)</code> pattern used in every agent               |
| <b>Website</b> | <a href="https://python.langchain.com/">https://python.langchain.com/</a> |

## 4. Web Search

### Tavily API

|                    |                                                                                                 |
|--------------------|-------------------------------------------------------------------------------------------------|
| <b>Role</b>        | Real-time web search — finds actual companies, decision makers, and contact details             |
| <b>Usage</b>       | Web Search Agent (parallel), Decision Maker Agent (per company), Enrichment Agent (per contact) |
| <b>Search mode</b> | basic depth, max 3–4 results per query                                                          |
| <b>Free Tier</b>   | 1,000 searches/month                                                                            |
| <b>Why chosen</b>  | LLM-optimised search API; returns clean snippets ready for prompt injection                     |
| <b>Website</b>     | <a href="https://tavily.com">https://tavily.com</a>                                             |

## 5. Backend

### FastAPI

|                   |                                                                                                 |
|-------------------|-------------------------------------------------------------------------------------------------|
| <b>Version</b>    | 0.115                                                                                           |
| <b>Role</b>       | REST API backend exposing the agent pipeline to the frontend                                    |
| <b>Endpoints</b>  | POST <code>/api/analyze</code> , POST <code>/api/approve</code> , POST <code>/api/report</code> |
| <b>CORS</b>       | Fully open ( <code>allow_origins=["*"]</code> ) for local development                           |
| <b>Server</b>     | Uvicorn 0.34.3 (ASGI)                                                                           |
| <b>Why chosen</b> | Fast, async, auto-generates OpenAPI docs; minimal boilerplate                                   |
| <b>Website</b>    | <a href="https://fastapi.tiangolo.com">https://fastapi.tiangolo.com</a>                         |

### Pydantic

|                   |                                                                          |
|-------------------|--------------------------------------------------------------------------|
| <b>Version</b>    | 2.11.4                                                                   |
| <b>Role</b>       | Request/response validation for all API endpoints                        |
| <b>Why chosen</b> | Native to FastAPI; enforces typed contracts between frontend and backend |

## 6. Frontend

### Streamlit

|            |                                                                          |
|------------|--------------------------------------------------------------------------|
| Version    | 1.45                                                                     |
| Role       | Interactive web UI — input form, 6-tab results dashboard, human approval |
| Port       | 8501                                                                     |
| Components | upload_section.py, results_section.py, approval_section.py               |
| Why chosen | Rapid UI development in pure Python; no HTML/JS required                 |
| Website    | <a href="https://streamlit.io">https://streamlit.io</a>                  |

## 7. PDF Report Generation

### PyMuPDF (fitz)

|            |                                                                             |
|------------|-----------------------------------------------------------------------------|
| Version    | 1.25.5                                                                      |
| Role       | Generates the 8-page professional PDF discovery report                      |
| Output     | prospectiq_report.pdf — branded, downloadable from the Streamlit UI         |
| Why chosen | Fast, precise PDF layout control with low-level drawing primitives          |
| Website    | <a href="https://pymupdf.readthedocs.io">https://pymupdf.readthedocs.io</a> |

## 8. Infrastructure & Configuration

### Python

|              |                                                                                          |
|--------------|------------------------------------------------------------------------------------------|
| Version      | 3.13                                                                                     |
| Role         | Core language for all agents, backend, and frontend                                      |
| Key features | TypedDict for typed state, ThreadPoolExecutor for parallel searches, asyncio via FastAPI |

### python-dotenv

|              |                                          |
|--------------|------------------------------------------|
| Role         | Loads API keys from .env file at startup |
| Keys managed | GROQ_API_KEY, TAVILY_API_KEY, GROQ_MODEL |

## 9. Summary Table

| Technology | Version | Layer | Purpose               |
|------------|---------|-------|-----------------------|
| Groq API   | —       | LLM   | LLM inference hosting |

|                |        |          |                                    |
|----------------|--------|----------|------------------------------------|
| LLaMA 3.1 8B   | —      | LLM      | Language model for all agents      |
| LangGraph      | 0.4.8  | Agents   | Multi-agent pipeline orchestration |
| LangChain      | 0.3.25 | Agents   | LLM integration layer              |
| Tavily API     | —      | Search   | Real-time web search               |
| FastAPI        | 0.115  | Backend  | REST API                           |
| Uvicorn        | 0.34.3 | Backend  | ASGI server                        |
| Pydantic       | 2.11.4 | Backend  | Data validation                    |
| Streamlit      | 1.45   | Frontend | Interactive UI                     |
| PyMuPDF (fitz) | 1.25.5 | Output   | PDF report generation              |
| Python         | 3.13   | Core     | Primary language                   |
| python-dotenv  | —      | Config   | Environment variable management    |